Xincheng Yao

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) **Guangzhou, China**
M.Phil. & Ph.D. in Data Science and Analytics *September 2024 - June 2029 (expected)*

– Supervisor: Lei Zhu

Zhejiang University **Hangzhou, China**
B.Sc. in Agriculture *September 2020 - June 2024*

– Supervisor: Yanhong Zhou

Key Courses: Data Mining, Artificial Intelligence for Medical Imaging, Physics-based Vision, Spatiotemporal Data Analysis, Data Science and Algorithms, Data Analysis and Machine Learning.

RESEARCH EXPERIENCE

Foundation Models and RL-Based Memory Scheduling for Long-Horizon Surgical Video Understanding — *Primary Contributor* *2024.11 - 2026.06*

- **Overview.** Developed HRVVS and VigilSAM3 for high-resolution perception and long-horizon memory drift in laparoscopic hepatectomy, targeting Glisson's pedicle, hepatic veins, and bleeding points; also contributed to the Hepa-SEG and Hepa-Vein datasets.
- **HRVVS.** Introduced pre-trained Visual Autoregressive (VAR) features as hierarchical residual priors, with adapter tuning, multi-view encoding, and MSIM/DWFM dynamic memory decoding to preserve fine vascular cues and improve temporal consistency.
- **VigilSAM3.** Built a hierarchical Memory Firewall and causal-Transformer agent on SAM3, modeling memory updates as Trust/Correct/Reset decisions; used SFT initialization and class-aware GRPO to mitigate ultra-long-video drift while reducing human intervention.

Ultrasound Video Segmentation and Multimodal Diagnosis with VLM-Derived Reasoning Priors — *Primary Contributor* *2024.09 - 2025.11*

- **Overview.** Developed MU-SAM2 for ultrasound video segmentation and HydroVLP for joint segmentation and grading of pediatric hydronephrosis, focusing on foundation-model adaptation and VLM-driven multimodal reasoning.
- **MU-SAM2.** Proposed two-stage SAM→SAM2 learning: multi-organ ultrasound montage pre-training learns general acoustic representations, while MLLM agents route organ-specific SAM teachers and distill expert priors into SAM2 through projected and reverse feature matching.
- **HydroVLP.** Used a LoRA-adapted VLM to generate frame semantics, keyframe signals, and grading rationales; combined task-driven keyframe selection, confidence-calibrated gating, and semantic-guided fusion over video, structured clinical tables, and language priors.

INTERNSHIP EXPERIENCE

Zhipu AI, Base Group **Beijing, China**
Post-training Intern *2026.5-Current*

- **Overview.** Developed long-horizon coding agents for iterative software development, screenshot-to-software generation, and editable scientific-diagram reconstruction, with verifiable workflows and reasoning traces for post-training.
- **RSI: Adaptive Long-Horizon Software Synthesis.** Designed a Meta–Worker–Evaluator framework for complexity-aware planning and evidence-driven replanning under frozen acceptance contracts; bound feedback to product versions and enforced regression gates to preserve completed capabilities across iterations.
- **Contract-Guided Cross-Platform Software Synthesis.** Introduced workflow and data contracts for Web-to-App transfer across web applications, shared backends, and Flutter code; used stage-isolated contexts

and structured summaries for long-horizon execution. One case study passed 30/31 browser tests.

- **LeanEditor: Direct Diagram-to-PPTX Compilation.** Compiled LLM-generated JavaScript directly into editable PPTX, bypassing intermediate formats to reduce conversion artifacts. Established, to our knowledge, the largest benchmark for editable scientific-diagram reconstruction; achieved pixel-level fidelity substantially surpassing prior state-of-the-art methods.

PUBLICATIONS

- **HRVVS: A High-resolution Video Vasculature Segmentation Network via Hierarchical Autoregressive Residual Priors**
  Xincheng Yao, Yijun Yang, Kangwei Guo, Ruiqiang Xiao, Haipeng Zhou, Haisu Tao, Jian Yang, and Lei Zhu
  International Conference on Medical Image Computing and Computer-Assisted Intervention (*MICCAI, CCF-B*), 2025.
- **VigilSAM3: Reinforcement-Optimized Memory Gating for Ultra-Long Surgical Video Segmentation**
  Xincheng Yao, Yijun Yang, Zhuojin Song, Kangwei Guo, Haisu Tao, Jian Yang, and Lei Zhu
  International Conference on Medical Image Computing and Computer-Assisted Intervention (*MICCAI, CCF-B*), 2026.
- **From Single Frames to Videos: Real-Time Anatomy Recognition in Laparoscopic Liver Resection Using Video Segmentation AI Model**
  Haisu Tao, Kangwei Guo, Yijun Yang, **Xincheng Yao**, Ruiqiang Xiao, Xiaojun Zeng, Xuanshuang Tang, Lei Zhu, Yinlin Qian, and Jian Yang
  npj Digital Medicine (*SCI Q1*), IF: 15.1, 2025.
- **HydroVLP: Multi-Task Learning with VLM-Derived Reasoning Priors for Ultrasound Hydronephrosis Diagnosis**
  Xincheng Yao, Yijun Yang, Ruicheng Ao, Yuandong Tao, and Lei Zhu
  IEEE Transactions on Medical Imaging (*TMI; SCI Q1*), IF: 12.4, 2026. (**Under Review**)
- **Universal to Specific: Distilling Multi-organ Ultrasound Knowledge into Segment Anything Model 2**
  Yijun Yang, **Xincheng Yao**, Ruiqiang Xiao, Xiaorui Liu, Lihao Liu, Huazhu Fu, Jing Qin, Huixia Zhou, Yuandong Tao, and Lei Zhu
  IEEE Transactions on Pattern Analysis and Machine Intelligence (*TPAMI; SCI Q1*), IF: 20.8, 2026. (**Under Review**)

RESEARCH INTERESTS

Recursive Self-Improvement System, Multi-agent Coding System, Medical Foundation Models

AWARDS & HONORS

- **First Prize, Chinese Mathematics Competitions for College Students**, 2023
- **Second Prize, Zhejiang University Mathematical Contest in Modeling**, 2023
- **Zhejiang University Scholarship for Academic Excellence**, 2022
- **Zhejiang University Third-Class Scholarship**, 2021

SKILLS

Programming Languages	Python, C#, LaTeX, Java, MATLAB
English Proficiency	IELTS: 6.5, CET-6: 552
Frameworks	PyTorch, PyTorch Lightning, Hugging Face Transformers